Scalar TM-free Feedback
Exact Supporting Convergence and Cycle Exclusion Results

Abstract

We study the scalar feedback subsystem of the printed TM-free algorithm, preserving its initialization, temporal-secant controller, and one-step extrapolation lag. The controller freezes after finitely many changes on every infinite defined strongly convex smooth trajectory. We give quadratic stability and exact rational exclusions of nonconstant period-three and period-four frozen or limiting cycles. For a relaxation retaining only consecutive secant bounds, the joint spectral radius is exactly $r\sqrt{1+2r}$. This yields a conditional convergence regime, while its extremal alternating mode violates scalar gradient interpolation. A separate rational function-value certificate proves geometric convergence at $m = 1/25$, $\mu = 1/50$, beyond the temporal-only stable regime. Complete proofs and exact certificate data are provided. General convergence from the printed initialization remains unresolved. The source leaves an output coefficient and zero-secant branch undefined; neither is silently completed here.

Keywords: adaptive acceleration; triple momentum; temporal secants; joint spectral radius; rational certificates

2020 Mathematics Subject Classification. Primary 65K10; Secondary 37N40, 39A30, 39A23.

1 Problem and precise scope

TM-free appears in Appendix C.3, Algorithm 2 of Cavalcanti, Lessard, and Wilson [1]. The convergence theorems for NAG-free in that work concern a different algorithm and do not establish TM-free convergence. The original constant-parameter triple-momentum theorem [2] assumes a valid global strong-convexity bound. A trajectory's secant estimate need not satisfy that assumption.

The target class consists of scalar C^1, strongly convex, piecewise-quadratic functions with finitely many breakpoints and curvatures in $[\mu, 1]$, where $0 < \mu < 1$. The minimizer is translated to zero. Several supporting theorems below hold in larger scalar smooth classes. The unresolved question is whether every infinite defined trajectory from the printed initialization converges in this target class, with controller factor $\eta = 3/2$. An initialized counterexample would have to verify its objective, controller branches, nonzero finite-time secants, and reachability from the prescribed initial state. A freely chosen frozen periodic state would not suffice. No higher-dimensional conclusion or novelty claim is made here.

1.1 The literal feedback subsystem

We use $\eta > 1$ for the controller factor denoted by γ in [1], and g_t for its coefficient γ_t. For algorithm input $L > 0$, initialization is exactly

$$\xi_{-1} = \xi_0 = y_0 = a, \qquad m_0 = L. \tag{1}$$

For $t \geq 0$, the printed feedback is

$$r_t = 1 - \sqrt{m_t/L}, \qquad \alpha_t = (1 + r_t)/L, \tag{2}$$

$$\beta_t = \frac{r_t^2}{2 - r_t}, \qquad g_t = \frac{r_t^2}{(1 + r_t)(2 - r_t)},$$

$$\xi_{t+1} = (1 + \beta_t)\xi_t - \beta_t \xi_{t-1} - \alpha_t f'(y_t), \tag{3}$$

$$y_{t+1} = (1 + g_t)\xi_{t+1} - g_t \xi_t, \tag{4}$$

$$c_{t+1} = \frac{|f'(y_{t+1}) - f'(y_t)|}{|y_{t+1} - y_t|}, \tag{5}$$

$$m_{t+1} = \begin{cases} \min\{m_t/\eta, c_{t+1}\}, & c_{t+1} < m_t, \\ m_t, & c_{t+1} \geq m_t. \end{cases} \tag{6}$$

Thus y_t uses g_{t-1} for $t \geq 1$ and is not recomputed with g_t at a switch. Algorithm 2 has no function-value acceptance test or backtracking step; its declared input γ_L is unused.

The source also prints $x_{t+1} = (1 + \delta_t)\xi_{t+1} - \delta_t \xi_t$ and returns x_T, but supplies no definition of δ_t. Original constant TM defines $\delta = r^2/(1-r^2)$ [2, Definition 2]; substituting r_t would be an additional completion. The undefined output does not feed into (3)–(6). All point-convergence results here concern ξ_t, y_t.

The quotient (5) is undefined when $y_{t+1} = y_t$, and the source gives no special branch. An *infinite defined literal trajectory* therefore means that every actually used quotient has nonzero denominator. Algebraic frozen-system results below may permit coincident samples; this does not define the missing controller branch. Starting at the minimizer already encounters 0/0.

Except in the preliminary statements retaining L, normalize $L = 1$. The parameter μ describes global objective curvature. The controller m_t is an estimate and is not assumed to be a valid global lower bound.

2 Finite freezing and quadratic stability

Lemma 1 (Finite controller freezing)**.** *For a scalar C^1, μ-strongly convex, L-smooth objective, where $0 < \mu \leq L$, every infinite defined literal trajectory has finitely many controller changes and final value*

$$\mu/\eta \leq m_\infty \leq L.$$

Neither boundedness of the trajectory nor a finite piecewise partition is needed.

Proof. For distinct u, v, scalar strong convexity and smoothness give

$$\mu \le \frac{f'(u) - f'(v)}{u - v} \le L.$$

Thus every defined c_{t+1} lies in $[\mu, L]$. A strict update occurs only while $m_t > \mu$ and reduces its value by at least the factor η. Infinitely many updates would force the pre-update values to zero, contradicting this requirement. At an update, both m_t/η and c_{t+1} are at least μ/η. This lower bound is preserved by unchanged steps, while m_t never increases from L. ∎

If $m_t = m$ for every $t \ge K$, the value y_K may still use the previous extrapolation coefficient. The frozen relation holds from y_{K+1}. For proofs involving both y_t, y_{t-1} and the previous update, it suffices to start at

$$t_0 = K + 2. \tag{7}$$

Discarding this finite prefix neither changes the actual trajectory nor affects its asymptotics.

Proposition 2.1 (Quadratic support)**.** Let $f(u) = \lambda u^2/2$, $0 < \lambda \le L$. The frozen feedback is Schur stable for every fixed controller $m \in (0, L]$. Hence every infinite defined literal trajectory on this quadratic converges to zero.

Proof. Put $r = 1 - \sqrt{m/L} \in [0, 1)$, $b = r^2/(2 - r)$, and $q = \lambda/L \in (0, 1]$. Substitution of the frozen y relation into the ξ update gives characteristic polynomial $z^2 - Az + C$, where

$$A = 1 + b - (1 + r + b)q, \qquad C = b(1 - q).$$

Its scalar Jury inequalities are

$$\begin{aligned} 0 \le C &< 1, \\ 1 - A + C &= (1 + r)q > 0, \\ 1 + A + C &= 2(1 + b) - (1 + r + 2b)q \ge 1 - r > 0. \end{aligned}$$

Both roots therefore lie strictly inside the unit disk. Lemma 1 supplies a positive frozen controller, and the finite prefix and coefficient lag do not change stability. ∎

Finite freezing alone is not nonlinear convergence: the final estimate can exceed the objective's global lower curvature, preventing direct use of the original fixed-parameter theorem.

3 Exact exclusion of short periodic orbits

For the rest of the report, frozen parameters have the form

$$m = (1 - r)^2, \quad \alpha = 1 + r, \quad b = \frac{r^2}{2 - r}, \quad g = \frac{b}{1 + r}, \qquad 0 \le r < 1. \tag{8}$$

A frozen periodic orbit satisfies both the ξ update and the frozen y relation. A temporal-secant constraint is imposed whenever its endpoints differ.

Theorem 1 (Periods three and four)**.** *Let $f : \mathbb{R} \to \mathbb{R}$ be C^1 with nondecreasing, 1-Lipschitz derivative. Any period-three or period-four orbit of the frozen feedback, whose nonzero temporal secants are at least m, is constant. Coincident consecutive or nonconsecutive evaluation points are allowed. The constant is a gradient root, and is zero if f is strongly convex with minimizer zero.*

3.1 Periodic interpolation inequalities

Assume $0 < r < 1$, $n \in \{3, 4\}$, and write $X = (\xi_0, \dots, \xi_{n-1})^\mathsf{T}$ with indices modulo n. Let $D = (1 + r)(2 - r) > 0$ and define row operators

$$Y_i X = (2 + r)\xi_i - r^2 \xi_{i-1}, \tag{9}$$

$$H_i X = (2 - r + r^2)\xi_i - r^2 \xi_{i-1} - (2 - r)\xi_{i+1}. \tag{10}$$

The frozen equations give exactly

$$Y_i X = D y_i, \qquad H_i X = D f'(y_i). \tag{11}$$

There is no free additive intercept in the gradient representation.

Choose a spatial order $p = (p_0, \dots, p_{n-1})$ with $y_{p_0} \le \cdots \le y_{p_{n-1}}$, rotate time to make $p_0 = 0$, and break ties arbitrarily. Construct $\mathcal{A}_{n,p}$ with zero-based row indices. For each adjacent spatial pair $i = p_k, j = p_{k+1}$ in order $k = 0, \dots, n - 2$, append the three rows

$$-(Y_j - Y_i), \qquad -(H_j - H_i), \qquad (H_j - H_i) - (Y_j - Y_i). \tag{12}$$

Next, for the temporal edges $i \to i + 1$ in order $i = 0, \dots, n - 1$, orient each pair (a_i, b_i) from lower to higher spatial rank and append

$$(1 - r)^2 (Y_{b_i} - Y_{a_i}) - (H_{b_i} - H_{a_i}). \tag{13}$$

Every admissible cycle then satisfies

$$\mathcal{A}_{n,p} X \le 0, \qquad R_{n,p} X \ge 0, \qquad R_{n,p} = Y_{p_{n-1}} - Y_{p_0}. \tag{14}$$

The rows in (12) impose ordering, monotonicity, and Lipschitzness. If spatial points coincide, their gradient values must coincide by the last two inequalities. A tied temporal edge contributes zero to (13). Thus ties remain admissible without assigning a quotient value.

3.2 Proof by exact rational certificates

Proof of Theorem 1. Appendix A supplies all ten rational certificates needed for the two period-three orders and six period-four orders. Each gives selected row indices $j_1, \dots, j_s$ and nonnegative weights $w_1, \dots, w_s$, $s \le n-1$, with exact identity

$$\sum_{\ell=1}^{s} w_\ell (\mathcal{A}_{n,p})_{j_\ell,:} = R_{n,p}. \tag{15}$$

Two period-four orders use two branches according to the sign of

$$P(r) = r^5 + r^4 + 3r^3 + 3r^2 - 4. \tag{16}$$

Both non-strict branches include $P = 0$, where the relevant weight vanishes and every denominator stays nonzero. The appendix gives the full rational formulas and sign proofs. Substituting (9)–(13) verifies every component of (15).

Applying (15) to (14) gives $R_{n,p}X \le 0$. Its reverse inequality is the ordering, so all y_i coincide, say at q. The cyclic operator $(2+r)I - r^2\mathsf{S}$, with backward shift S, is invertible since $r^2/(2+r) < 1$. Its solution for right-hand side $Dq\mathbf{1}$ is $X = q\mathbf{1}$, because $2+r-r^2 = D$. The frozen update now implies $f'(q) = 0$.

At $r = 0$, the method is $y = \xi$, $T(u) = u - f'(u)$. For $u > v$, $0 \le T(u) - T(v) \le u - v$, so T is nondecreasing. Such a map has no nonconstant finite cycle: if $T(u) > u$, all later iterates are at least $T(u) > u$, while if $T(u) < u$ the reverse holds. A periodic orbit must therefore be fixed. Strong convexity makes its gradient root unique when assumed. ∎

Corollary 3.1 (Literal tails and limiting cycles)**.** *An infinite defined literal trajectory on a scalar strongly convex* 1*-smooth objective has no nontrivial eventually periodic tail of period at most four and cannot converge phase by phase to a nontrivial cycle of period at most four.*

Proof. After finite freezing and the lag prefix, the recurrence is frozen and obeys the division-free inequality

$$(f'(y_{t+1}) - f'(y_t))(y_{t+1} - y_t) \ge m(y_{t+1} - y_t)^2. \tag{17}$$

Theorem 1 excludes eventually periodic tails; period two embeds in period four, and a fixed point is a gradient root.

If the full ξ/y state converges on each residue class modulo $n \le 4$, continuity passes the frozen recurrence and (17) to the limit. Monotonicity and the upper Lipschitz inequality pass as well. Nonzero limiting differences recover the secant lower bound. Zero limiting differences have equal limiting gradients. The same certificates therefore apply even if a limiting quotient would be undefined. This proves the conclusion without asserting boundedness or existence of a periodic limit set for a general trajectory. ∎

Remark. The excluded endpoint $r = 1$, $m = 0$, cannot be added: for $f(y) = y^2/2$, parameters $\alpha = 2, b = 1, g = 1/2$ admit $\xi_i = (-1)^i a$, $y_i = 2(-1)^i a$ for every $a \neq 0$.

There is also a second period-three argument. The three temporal edges cover every pair. After merging ties, their gradient samples admit a piecewise-linear interpolant with slopes in $[m, 1]$. Extend the outer segments with slopes in the same interval and integrate. The resulting globally m-strongly convex, 1-smooth objective has the same periodic samples, contradicting constant-TM convergence, including ξ convergence in [2, Corollary 1]. The rational proof above does not depend on importing any adaptive output definition.

4 The exact temporal-secant stability boundary

The temporal-only relaxation retains consecutive slope bounds but forgets whether all samples arise from a single objective.

Lemma 2 (Difference dynamics)**.** *For $0 < r < 1$, let $v_t = \xi_t - \xi_{t-1}$. Sufficiently after freezing,*

$$\begin{pmatrix} v_{t+1} \\ v_t \end{pmatrix} = M(c_t) \begin{pmatrix} v_t \\ v_{t-1} \end{pmatrix}, \qquad M(c) = \begin{pmatrix} 1 + b - (1 + r + b)c & -b(1 - c) \\ 1 & 0 \end{pmatrix}, \tag{18}$$

where actual scalar secants satisfy $m \leq c_t \leq 1$.

Proof. Subtract consecutive frozen updates and extrapolations:

$$\begin{aligned} v_{t+1} &= (1 + b)v_t - bv_{t-1} - \alpha(f'(y_t) - f'(y_{t-1})), \\ y_t - y_{t-1} &= (1 + g)v_t - gv_{t-1}. \end{aligned}$$

An increasing scalar gradient makes the norm quotient equal the signed secant. Substitute $f'(y_t) - f'(y_{t-1}) = c_t(y_t - y_{t-1})$, and use $\alpha g = b$, $\alpha(1 + g) = 1 + r + b$. The controller and smoothness give the lower and upper bounds. Starting at (7) handles the lag. ∎

For a bounded family $\mathcal{M}$, write

$$\widehat{\rho}(\mathcal{M}) = \limsup_{n \to \infty} \sup_{M_i \in \mathcal{M}} \|M_n \cdots M_1\|^{1/n}$$

for its joint spectral radius, using any fixed induced norm.

Theorem 2 (Exact joint spectral radius)**.** *For $0 < r < 1$, $m = (1 - r)^2$, and $\mathcal{M}_r = \{M(c) : m \leq c \leq 1\}$,*

$$\widehat{\rho}(\mathcal{M}_r) = \Theta(r) := r\sqrt{1 + 2r}. \tag{19}$$

There is $C_r < \infty$ such that every length-n product has norm at most $C_r\Theta^n$. Arbitrary switching is uniformly exponentially stable exactly when $r^2(1 + 2r) < 1$. At equality all products are uniformly bounded but asymptotic stability fails. Above equality some switched trajectories are unbounded.

Proof. The endpoint matrices are

$$A = M(m) = \begin{pmatrix} r + r^2 & -r^3 \\ 1 & 0 \end{pmatrix}, \quad B = M(1) = \begin{pmatrix} -r & 0 \\ 1 & 0 \end{pmatrix} = ue_1^{\mathsf{T}}, \quad u = \begin{pmatrix} -r \\ 1 \end{pmatrix}. \tag{20}$$

The eigenvalues of A are the distinct numbers r, r^2, so $\|A^j\| \le C_A r^j$ for all $j \ge 0$. Let

$$\tau_k = e_1^{\mathsf{T}} A^k u, \qquad S_k = 1 + 2\sum_{i=1}^{k} r^i.$$

The characteristic recurrence gives

$$\tau_k = -r^{k+1} S_k. \tag{21}$$

Indeed both sides have initial values $-r, -r^2(1+2r)$ and satisfy $\tau_{k+2} = (r+r^2)\tau_{k+1} - r^3\tau_k$.

Put $s = 1 + 2r$. We prove $S_k \le s^{(k+1)/2}$. The case $k = 0$ is immediate and $k = 1$ is equality. For $k = 2$,

$$s^3 - S_2^2 = 2r + 4r^2 - 4r^4 > 0.$$

For $k \ge 1$,

$$sS_k - S_{k+2} = 2r[S_k - r^k(1 + r)],$$
$$S_k - r^k(1 + r) = 1 + 2\sum_{i=1}^{k-1} r^i + (1 - r)r^k > 0.$$

Two-step induction covers both parities. Thus $|\tau_k| \le \Theta^{k+1}$ for every $k \ge 0$.

An endpoint word with $q \ge 1$ copies of B factors as

$$W = A^a B A^{k_{q-1}} B \cdots B A^{k_1} B A^d$$
$$= A^a u \left(\prod_{i=1}^{q-1} \tau_{k_i} \right) e_1^{\mathsf{T}} A^d,$$

where $a, d, k_i \ge 0$ and $n = a + d + q + \sum_i k_i$ is its length. Therefore

$$\|W\| \le C_A^2 \|u\| \, r^{a+d} \Theta^{n-a-d-1} \le \frac{C_A^2 \|u\|}{\Theta} \Theta^n.$$

This includes adjacent B factors, $q = 1$, and incomplete outer A blocks. Words without B satisfy $\|A^n\| \le C_A \Theta^n$.

Each $M(c)$ is a convex combination of A, B. Expanding a product gives a convex combination of endpoint words of the same length, proving the claimed uniform bound. Conversely, BA has eigenvalues 0 and $e_1^{\mathsf{T}} Au = -r^2(1 + 2r) = -\Theta^2$. Repeating this word attains the growth factor Θ per step, giving equality in (19). At $\Theta = 1$, $(BA)^j u = (-1)^j u$ does not tend to zero despite the uniform bound. At $\Theta > 1$ the same eigenmode is unbounded. ∎

Corollary 4.1 (A sufficient actual convergence regime)**.** *Let r_* be the unique root in $(0,1)$ of $2r^3+r^2-1=0$, and $m_*=(1-r_*)^2\approx 0.1174445881$. Let $f:\mathbb{R}\to\mathbb{R}$ be C^1, strongly convex and globally 1-smooth, with minimizer zero. If an infinite defined literal trajectory freezes at $m\in(m_*,1)$, then*

$$|\xi_t|+|y_t|=O(\Theta(1-\sqrt{m})^t).$$

This does not require m to be a valid global curvature bound.

Proof. The polynomial defining r_* is strictly increasing on $(0,1)$ and changes sign there. The hypothesis is exactly $\Theta<1$. The matrix bound gives $|v_t|\le C\Theta^t$, absorbing the finite prefix in C. Thus $\sum_t|v_t|<\infty$, $\xi_t\to\ell$ for some finite ℓ, and $y_t=\xi_t+gv_t\to\ell$. The actual update implies

$$\alpha f'(y_t)=bv_t-v_{t+1}\longrightarrow 0.$$

Continuity and strong convexity give $f'(\ell)=0$ and $\ell=0$. Summing the geometric tail gives the stated point rate. ∎

The printed initialization does not force $m_\infty>m_*$. At $m=1$, the frozen feedback is gradient descent and a frozen temporal secant would have to equal 1, forcing a subsequent difference to vanish. Hence an infinite tail with every literal quotient defined cannot persist at that exact endpoint. No zero-secant completion is asserted.

Proposition 4.1 (A Lyapunov obstruction for the relaxation)**.** If $r^2(1+2r)>1$, there is no finite-valued common nonincreasing coercive Lyapunov function for the arbitrary-switch family. The same holds for a fixed finite-history function coercive in the current difference state and required to decrease on every admissible switching history.

Proof. The alternating endpoint word has an unbounded trajectory. A nonincreasing function would confine it to the sublevel set of its finite initial value; coercivity makes that set bounded in the current state. For a history function choose a finite admissible alternating initial history and repeat the same argument. ∎

This obstruction does not rule out a potential exploiting nonadjacent interpolation or function values.

5 The extremal switching mode is not interpolable

Proposition 5.1. For every $0<r<1$, the nonzero eigenmode of BA cannot be generated by a single nondecreasing 1-Lipschitz scalar gradient. It therefore supplies no optimization counterexample.

Proof. Let $K=r^2(1+2r)$, $H=r(1+r)$, $D_0=2r/(2-r)>0$, and start the formal difference state at $(v_0,v_{-1})=(-r,1)$. Take formal secants $c_0=m, c_1=1, c_2=m$. Then $v_1=-K$, $v_2=rK$, and

$$d_0=-D_0,\quad d_1=-D_0H,\quad d_2=D_0K,\qquad d_t=(1+g)v_t-gv_{t-1}.$$

Define formal gradients by $G_t - G_{t-1} = c_t d_t$. The nonadjacent pair $0, 2$ has

$$y_2 - y_0 = D_0(K - H), \qquad G_2 - G_0 = D_0(mK - H). \tag{22}$$

We have $K - H = r(2r^2 - 1)$ and $mK < H$, since

$$(1 - r)^2(1 + 2r) = 1 - 3r^2 + 2r^3 < 1$$

implies $mK < r^2 < H$.

If $r > 1/\sqrt{2}$, the point difference in (22) is positive and the gradient difference negative, violating monotonicity. If $r < 1/\sqrt{2}$, the pair's secant exceeds one because

$$\frac{mK - H}{K - H} - 1 = \frac{(m - 1)K}{K - H} > 0.$$

At $r = 1/\sqrt{2}$, the points coincide but gradients differ, violating single-valuedness. Translations and nonzero scalings preserve these contradictions. ∎

The exact matrix threshold is necessary only for the relaxation. It is a sufficient threshold for the actual feedback, not a necessary one.

6 A rational interpolation energy at one parameter pair

Theorem 3 (Fixed-controller convergence certificate)**.** *Let $f : \mathbb{R} \to \mathbb{R}$ be C^1, globally 1-smooth and $(1/50)$-strongly convex, with minimizer x_*. Suppose an infinite tail obeys*

$$\begin{aligned} \xi_{t+1} &= \frac{23}{15}\xi_t - \frac{8}{15}\xi_{t-1} - \frac{9}{5}f'(y_t), \\ y_t &= \frac{35}{27}\xi_t - \frac{8}{27}\xi_{t-1}, \end{aligned} \tag{23}$$

and its consecutive samples satisfy

$$\begin{aligned} &\left[f'(y_t) - f'(y_{t-1}) - \frac{1}{25}(y_t - y_{t-1})\right] \\ &\qquad \cdot [(y_t - y_{t-1}) - (f'(y_t) - f'(y_{t-1}))] \geq 0. \end{aligned} \tag{24}$$

Then $\xi_t, y_t \to x_*$ geometrically. An explicit positive energy contracts by

$$q_E = 1 - \frac{1}{40000000} \tag{25}$$

per frozen step. In particular, this applies to every infinite defined literal trajectory freezing at $m = 1/25$, under the stated global curvature bounds. It does not assume that m is a valid global curvature bound.

The exact coefficients correspond to

$$r = \frac{4}{5}, \quad m = \frac{1}{25}, \quad \alpha = \frac{9}{5}, \quad b = \frac{8}{15}, \quad g = \frac{8}{27}, \qquad \mu = \frac{1}{50}. \tag{26}$$

Here $\Theta^2 = r^2(1 + 2r) = 208/125 > 1$, outside the temporal-only stable regime. Function-value interpolation supplies information missing from that relaxation. The conservative factor (25) is not an accelerated-rate claim.

6.1 Interpolation inequality

Lemma 3 (Smooth strongly convex interpolation)**.** *For a scalar C^1, μ-strongly convex, 1-smooth function, $0 < \mu < 1$, any sampled data $(q_i, h_i, F_i) = (q_i, f'(q_i), f(q_i))$ satisfy*

$$H_{ij} := F_i - F_j - h_j(q_i - q_j) \\ - \frac{(h_i - h_j)^2 - 2\mu(h_i - h_j)(q_i - q_j) + \mu(q_i - q_j)^2}{2(1-\mu)} \geq 0. \tag{27}$$

Proof. Put $\phi(q) = f(q) - \mu q^2/2$. For $u > v$, the assumptions imply

$$0 \leq \phi'(u) - \phi'(v) \leq (1-\mu)(u - v),$$

so ϕ is convex and $(1-\mu)$-smooth. Fix q_j and define $\psi(q) = \phi(q) - \phi'(q_j)q$, which is minimized at q_j. The smooth descent bound with $L_\phi = 1 - \mu$ gives

$$\psi\left(q_i - \frac{\psi'(q_i)}{L_\phi}\right) \leq \psi(q_i) - \frac{|\psi'(q_i)|^2}{2L_\phi}.$$

Since $\psi(q_j)$ is no larger than the left side,

$$\phi(q_i) - \phi(q_j) - \phi'(q_j)(q_i - q_j) \geq \frac{|\phi'(q_i) - \phi'(q_j)|^2}{2(1-\mu)}.$$

Substitution of $\phi = f - \mu q^2/2$ and expansion yield (27). No second derivative or piecewise partition is needed. ∎

6.2 State reconstruction and the exact identity

Translate x_* to zero and subtract $f(x_*)$. This preserves the recurrence because the two ξ coefficients sum to one. Thus $f(0) = f'(0) = 0$. At a time sufficiently inside the frozen tail write

$$x = \xi_t, \quad p = \xi_{t-1}, \quad v = f'(y_{t-1}), \quad u = f'(y_t), \quad z = (x, p, v)^\mathsf{T}, \quad w = (x, p, v, u)^\mathsf{T}.$$

The previous and current evaluation points are

$$a = y_{t-1} = \frac{5}{9}x + \frac{4}{9}p + v, \qquad b_0 = y_t = \frac{35}{27}x - \frac{8}{27}p. \tag{28}$$

The symbol b_0 distinguishes the sample from the momentum coefficient b. To verify the previous sample, solve the actual previous update for

$$\xi_{t-2} = \frac{(1+b)p - x - \alpha v}{b}$$

and substitute into $y_{t-1} = (1+g)p - g\xi_{t-2}$. In particular the coefficient of v is $\alpha g/b = 1$. This is elimination along the actual dynamics, not an alternative iteration.

Define

$$\mathsf{A} = \begin{pmatrix} 23/15 & -8/15 & 0 & -9/5 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \qquad \mathsf{B} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}. \tag{29}$$

Then $z_t = \mathsf{B}w$, $z_{t+1} = \mathsf{A}w$. Let $D_E = 10^9$ and set

$$P_E = \frac{1}{D_E} \begin{pmatrix} 528430321 & -435784951 & 283032189 \\ -435784951 & 359466069 & -233442084 \\ 283032189 & -233442084 & 151618592 \end{pmatrix}, \tag{30}$$

$$\lambda = \frac{255107146}{D_E}, \quad \tau = \frac{398410159}{D_E}, \qquad V_t = z_t^{\mathsf{T}} P_E z_t + f(y_{t-1}). \tag{31}$$

Use the three samples $o = (0,0,0)$, $a = (a, v, f(a))$, and $b_0 = (b_0, u, f(b_0))$ in (27). With

$$\mathcal{D}(h,q) = \frac{h^2 - 2\mu hq + \mu q^2}{2(1-\mu)},$$

the active directed inequalities are explicitly

$$\begin{aligned} H_{ob_0} &= -f(b_0) + ub_0 - \mathcal{D}(u, b_0), \\ H_{ao} &= f(a) - \mathcal{D}(v, a), \\ H_{ab_0} &= f(a) - f(b_0) - u(a - b_0) - \mathcal{D}(v - u, a - b_0). \end{aligned} \tag{32}$$

The temporal inequality is

$$T = (u - v - m(b_0 - a))\big((b_0 - a) - (u - v)\big) \geq 0. \tag{33}$$

For a defined literal frozen step, $T = (c-m)(1-c)(b_0-a)^2$ with $c \in [m, 1]$. For coincident evaluation points, actual gradients coincide and $T = 0$. The algebraic theorem permits this equality, but it does not complete the literal quotient.

The exact polynomial identity is

$$\begin{aligned} V_t - V_{t+1} = w^{\mathsf{T}} S_E w + \lambda H_{ob_0} + \lambda H_{ao} \\ + (1 - \lambda) H_{ab_0} + \tau T, \end{aligned} \tag{34}$$

where Appendix B prints every entry of S_E. For an unambiguous construction, write $H_{ij} = F_i - F_j + w^{\mathsf{T}} Q_{ij} w$ and $T = w^{\mathsf{T}} Q_T w$ using (28) and (32). Then

$$\begin{aligned} S_E = \mathsf{B}^{\mathsf{T}} P_E \mathsf{B} - \mathsf{A}^{\mathsf{T}} P_E \mathsf{A} - \lambda Q_{ob_0} - \lambda Q_{ao} \\ - (1 - \lambda) Q_{ab_0} - \tau Q_T. \end{aligned} \tag{35}$$

The function values on the right of (34) have total coefficients 1 on $f(a)$ and -1 on $f(b_0)$, as in $V_t - V_{t+1}$. The remaining quadratic identity is exactly (35). No residual is unspecified.

6.3 Exact positivity and actual point convergence

Proof of Theorem 3. The rational principal minors in Appendix B prove by Sylvester's criterion that

$$P_E \succ \varepsilon I_3, \qquad S_E \succ \varepsilon I_4, \qquad \varepsilon = \frac{1}{20000000}. \tag{36}$$

All interpolation weights and τ are positive. Since $f(a) \geq 0$, Lemma 3, (33), and the exact identity give

$$V_t \geq \varepsilon \|z_t\|^2, \qquad V_t - V_{t+1} \geq \varepsilon \|w_t\|^2 \geq \varepsilon \|z_t\|^2. \tag{37}$$

Smoothness gives $f(a) \leq a^2/2$. For $k = (5/9, 4/9, 1)^{\mathsf{T}}$, we have $a = k^{\mathsf{T}} z$ and $\|k\|^2 = 122/81$. Positive definiteness of P_E yields

$$V_t \leq \left(\operatorname{tr}(P_E) + \frac{1}{2}\|k\|^2 \right) \|z_t\|^2,$$
$$\operatorname{tr}(P_E) + \frac{1}{2}\|k\|^2 = \frac{72600356771}{40500000000} < 2. \tag{38}$$

Consequently $V_{t+1} \leq (1 - \varepsilon/2) V_t = q_E V_t$. For a valid starting index t_0,

$$\|z_t\| \leq \sqrt{V_{t_0}/\varepsilon}\, q_E^{(t-t_0)/2}. \tag{39}$$

This bounds $|\xi_t|$; also $|y_t| \leq \sqrt{1289}\|z_t\|/27$ by (23). Undoing translation proves geometric convergence to x_*. Alternatively, summing (37) gives $\sum_t \|w_t\|^2 < \infty$ and hence actual point convergence directly.

For the literal statement, choose K so that $m_t = 1/25$ for $t \geq K$, and begin at $t_0 = K + 2$. Both evaluation formulas and the previous update then use the frozen coefficients, and the temporal secant lies in $[1/25, 1]$. The proof applies to the actual tail, whatever finite initialization prefix led to it, without assuming a freely chosen state is reachable. ∎

The theorem includes objectives whose best global lower curvature is exactly $1/50 < 1/25$. It does not show that any particular source-initialized objective reaches the exact final controller $1/25$. This is a conditional implication at one parameter pair.

7 Conclusions and remaining limits

Finite freezing and quadratic stability explain why controller overestimation alone does not imply nonlinear failure. The rational certificates exclude short frozen cycles and their phasewise limits. The exact joint spectral radius completely resolves the temporal-only matrix relaxation, but its extremal mode is not interpolable. The function-value energy at $(m, \mu) = (1/25, 1/50)$ supplies a certified convergence case beyond that relaxation's stable regime.

General initialized convergence remains unresolved, even for the finite-piece scalar target class. No exact exclusion of all periods five or larger, initialized counterexample, reachability theorem, or basin certificate is established. Numerical infeasibility screens are not exact nonexistence proofs. The energy theorem is restricted to its certified parameter pair; no open parameter box or whole-parameter extension is asserted. There is no conclusion about the undefined printed output, an invented zero-secant branch, higher-dimensional feedback, or a nonlinear accelerated rate.

A Complete rational short-cycle certificates

All row indices below are zero-based and refer exactly to (12)–(13). Define

$$\begin{aligned} Q(r) &= r^5 + 4r^4 + r^3 + 4r^2 + 4r + 4, \\ R(r) &= 2 + r - r^3, \qquad U(r) = r^4 + r^2 + 4r + 4, \\ N(r) &= r^7 + 3r^6 + r^5 + 3r^4 + 4r^2 + 4r + 8, \end{aligned}$$

and retain $P(r)$ from (16). The complete set of weights is

$$a_1 = \frac{r^2 - r + 2}{(2-r)(1-r)}, \qquad a_2 = \frac{2r}{(2-r)(1-r)(r^2+r+1)}, \tag{40}$$

$$b_1 = \frac{2r^2(r+1)(r^2+r+1)}{(1-r)Q(r)}, \qquad b_2 = \frac{N(r)}{(2-r)(1-r)Q(r)}, \tag{41}$$

$$b_3 = \frac{2r(r+1)(r^2-2r+4)}{(2-r)(1-r)Q(r)},$$

$$c_1 = \frac{r^4 - r^3 + r^2 + r + 2}{(2-r)(1-r)(r+1)^2}, \qquad c_2 = \frac{2r}{(2-r)(1-r)(r+1)^2}, \tag{42}$$

$$d_1 = \frac{r^2 + r + 2}{(2-r)(1-r)}, \qquad d_2 = \frac{2r^2(r+1)}{(1-r)R(r)}, \tag{43}$$

$$d_3 = \frac{-P(r)}{(2-r)(1-r)R(r)},$$

$$e_1 = \frac{P(r)}{(1-r)U(r)}, \qquad e_2 = \frac{2r(r+1)(r+2)}{(1-r)U(r)}, \tag{44}$$

$$f_1 = \frac{(r+2)(r^3-r+2)}{2(2-r)(1-r)(r^2+r+2)}, \qquad f_2 = \frac{2+3r+2r^2-r^3}{2(1-r)(r^2+r+2)}. \tag{45}$$

The symbols e_1, e_2 here are scalar weights, not coordinate vectors.

n	Spatial order p	Selected rows	Weights	Condition
3	$(0,1,2)$	$(5,6)$	(a_1,a_2)	all r
3	$(0,2,1)$	$(2,7)$	(a_1,a_2)	all r
4	$(0,1,2,3)$	$(1,8,10)$	(b_1,b_2,b_3)	all r
4	$(0,1,3,2)$	$(5,9,11)$	(c_1,c_2,c_2)	all r
4	$(0,2,1,3)$	$(5,6,8)$	(d_1,d_2,d_3)	$P \le 0$
4	$(0,2,1,3)$	$(0,5,6)$	(e_1,d_1,e_2)	$P \ge 0$
4	$(0,2,3,1)$	$(0,2,5)$	(d_2,d_3,d_1)	$P \le 0$
4	$(0,2,3,1)$	$(0,5,6)$	(e_2,d_1,e_1)	$P \ge 0$
4	$(0,3,1,2)$	$(2,4,8)$	(f_1,f_2,f_1)	all r
4	$(0,3,2,1)$	$(2,7,11)$	(b_2,b_1,b_3)	all r

For every row, the displayed weighted sum equals $Y_{p_{n-1}} - Y_{p_0}$ identically as a row vector. For example, $n=3$, $p=(0,1,2)$ gives

$$\begin{aligned} &a_1[(H_2-H_1)-(Y_2-Y_1)] \\ &\quad + a_2[(1-r)^2(Y_1-Y_0)-(H_1-H_0)] = Y_2-Y_0. \end{aligned} \tag{46}$$

The remaining identities have the same meaning, with all rows and coefficients specified. Clearing denominators reduces each component to the zero polynomial. The supplement verifies them by exact arithmetic from two independent reconstructions, one using the row formulas and one the original recurrence.

All weights have elementary signs throughout $0 < r < 1$. Every denominator is positive as written: $1-r$, $2-r$, $r+1$, r^2+r+1, and r^2+r+2 are positive; Q, U, N have positive coefficients, and $R = 2+r(1-r^2) > 2$. The non-immediate numerator signs follow from

$$\begin{aligned} r^2 - r + 2 &= (r-1/2)^2 + 7/4 > 0, \\ r^2 - 2r + 4 &= (r-1)^2 + 3 > 0, \\ r^4 - r^3 + r^2 + r + 2 &= r^2(1-r+r^2) + r + 2 > 0, \\ r^3 - r + 2 &> 1, \\ 2 + 3r + 2r^2 - r^3 &= 2 + 3r + r^2(2-r) > 0. \end{aligned}$$

Hence every weight except d_3, e_1 is strictly positive. The weight d_3 is nonnegative on $P \le 0$ and e_1 on $P \ge 0$. No denominator involves P. Also

$$P(0) = -4, \quad P(1) = 4, \quad P'(r) = 5r^4 + 4r^3 + 9r^2 + 6r > 0,$$

so the sole interior sign change is covered by both non-strict branches.

Every weak spatial order admits a linear extension after rotating a minimum to index zero, and the table contains all $(n-1)!$ such permutations. Thus ties are covered without a limiting perturbation argument. The independent exact verifier also enumerates all 13 weak orders for three samples and all 75 for four samples.

B Complete rational energy data

The residual in (34) is $S_E = K_E/33075000000000$, where

$$K_E = \begin{pmatrix} 9296245029857 & -8246820368257 & 14046593121015 & -4305893800903 \\ -8246820368257 & 7369593030207 & -12406400398140 & 3876410092228 \\ 14046593121015 & -12406400398140 & 21552284930400 & -6325159684284 \\ -4305893800903 & 3876410092228 & -6325159684284 & 2110095333432 \end{pmatrix}.$$

Together with (30), (31), and (32), this specifies the full identity in the four independent state variables and two independent function-value symbols.

Let $\Delta_j(M)$ denote the determinant of the upper-left $j \times j$ block. For $\varepsilon = 1/20000000$, the shifted leading minors are

$$\begin{aligned}
\Delta_1(P_E - \varepsilon I_3) &= \frac{528430271}{1000000000}, \\
\Delta_2(P_E - \varepsilon I_3) &= \frac{11050579347187}{250000000000000000}, \\
\Delta_3(P_E - \varepsilon I_3) &= \frac{527112949349724693}{100000000000000000000000000}, \\
\Delta_1(S_E - \varepsilon I_4) &= \frac{9296243376107}{33075000000000}, \\
\Delta_2(S_E - \varepsilon I_4) &= \frac{1427053804871810808271}{3125587500000000000000000}, \\
\Delta_3(S_E - \varepsilon I_4) &= \frac{2352386315062045610301434 3791}{625117500000000000000000000000000}, \\
\Delta_4(S_E - \varepsilon I_4) &= \frac{3553690739424227784644131 48210763}{976746093750000000000000000000000000000000000}.
\end{aligned}$$

Every numerator and denominator is a positive integer. Sylvester's criterion proves (36) exactly, without rounded eigenvalues. For an additional check, the unshifted minors are

$$\begin{aligned}
\Delta_1(P_E) &= \frac{528430321}{1000000000}, \\
\Delta_2(P_E) &= \frac{11061678051437}{250000000000000000}, \\
\Delta_3(P_E) &= \frac{530281381237087243}{100000000000000000000000000}, \\
\Delta_1(S_E) &= \frac{9296245029857}{33075000000000}, \\
\Delta_2(S_E) &= \frac{1427132550948830641921}{3125587500000000000000000}, \\
\Delta_3(S_E) &= \frac{2352628051476837146828 3872541}{625117500000000000000000000000000}, \\
\Delta_4(S_E) &= \frac{4454934634175425873191 7895328193}{610466308593750000000000000000000000000000000}.
\end{aligned}$$

C Reproducibility

The accompanying certificate supplement uses Python and SymPy. Periodic-orbit data are rational functions of the symbolic parameter; energy data are rational numbers. No numerical optimizer, trajectory search, or stored floating-point matrix is needed.

The cycle verifier reconstructs the inequalities, checks all rational identities, proves signs over the open unit interval, and checks every spatial permutation. An independent implementation reconstructs gradients from the scalar recurrence, verifies signs after $r = t/(1+t)$ for $t > 0$, and checks weak orders and the excluded $r = 1$ example. The energy verifiers independently recover the previous sample and residual matrix, check rational principal minors and strict margins, and verify the contraction constant. The temporal verifier checks the endpoint matrices and displayed algebra. Its finite symbolic checks supplement the analytical all-length proof of Theorem 2.

References

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